Chatbot setup using AZURE AI SERVICE:

NGA Artificial Intelligence

Exported on 09/03/2024

Table of Contents

1 What is Azure OpenAI Service? 3
2 What data does the Azure OpenAI Service process? 4
3 video by Devdas on creation of Azure AI chatbot based on your documents 5

1 What is Azure OpenAI Service?

Azure OpenAI Service provides REST API access to OpenAI's powerful language models including the GPT-4, GPT-4 Turbo with Vision, GPT-3.5-Turbo, and Embeddings model series. In addition, the new GPT-4 and GPT-3.5-Turbo model series have now reached general availability. These models can be easily adapted to your specific task including but not limited to content generation, summarization, image understanding, semantic search, and natural language to code translation. Users can access the service through REST APIs, Python SDK, or our web-based interface in the Azure OpenAI Studio.

2 What data does the Azure OpenAI Service process?

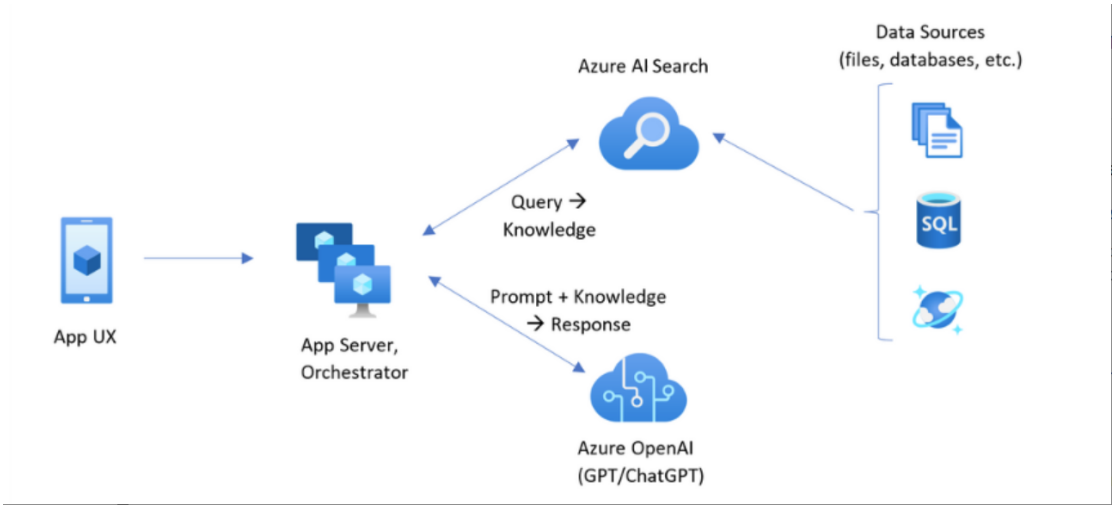
- **Azure OpenAI processes the following types of data:**

Prompts and generated content. Prompts are submitted by the user, and content is generated by the service, via the completions, chat completions, images and embeddings operations.

Augmented data included with prompts. When using the "on your data" feature, the service retrieves relevant data from a configured data store and augments the prompt to produce generations that are grounded with your data.

Training & validation data. You can provide your own training data consisting of prompt-completion pairs for the purposes of fine-tuning an OpenAI model[1].

ARCHITECTURE for RAG application in AZURE AI:



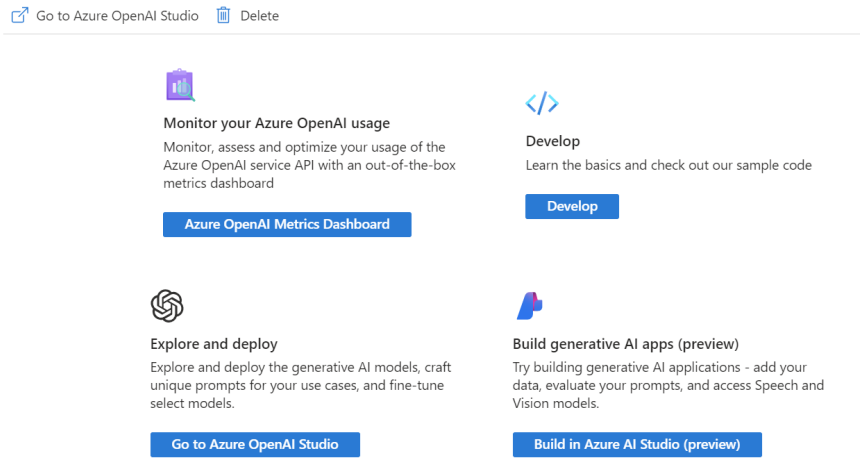
- **Options in Azure OpenAI:**

For more information follow this link-How-to: Create and deploy an Azure OpenAI Service resource - Azure OpenAI | Microsoft Learn[2]

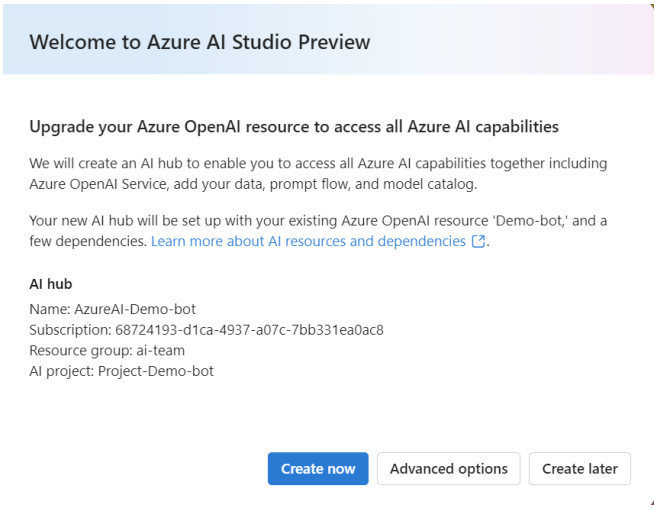
1 https://learn.microsoft.com/en-us/azure/ai-services/openai/how-to/fine-tuning?pivots=programming-language-studio
2 https://learn.microsoft.com/en-us/azure/ai-services/openai/how-to/create-resource?pivots=web-portal

3 video by Devdas on creation of Azure AI chatbot based on your documents[3]

Click on Build generative AI apps.



Select BUILD GENERATIVE AI APPS to build app using build in azure AI studio.



- **Available models in the Azure Open AI-**

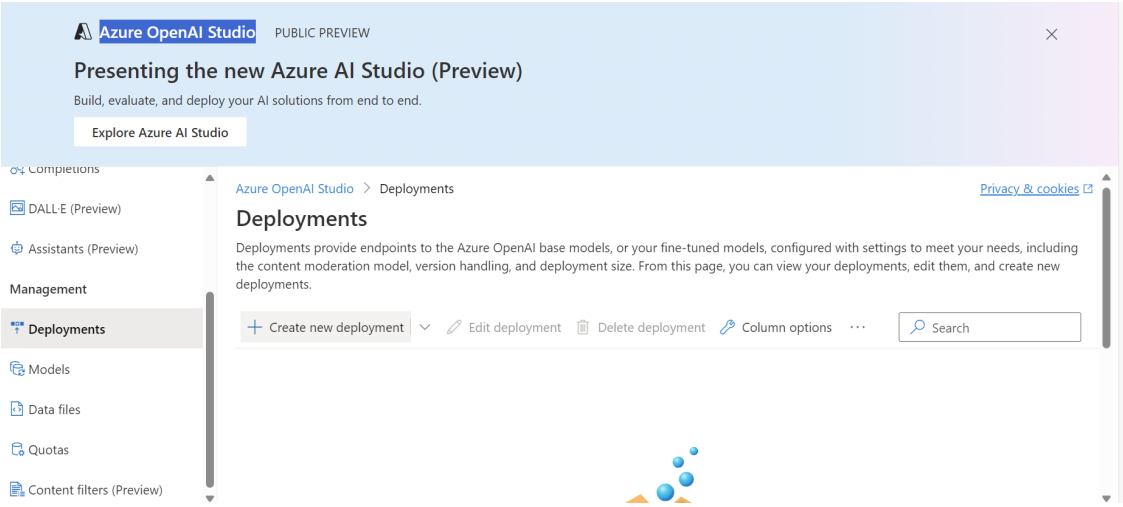
3 https://egonline-my.sharepoint.com/personal/presa_eg_dk/_layouts/15/stream.aspx?id=%2Fpersonal%2Fpresa%5Feg%5Fdk%2FDocuments%2FRecordings%2FCall%20with%20Devdas%20and%201%20other%2D20240216%5F102402%2DMeeting%20Recording%2FExports%2Fazure%5Fai%5Fchatbot%5Fsetup%5Fdemo%2Emp4&referrer=StreamWebApp%2EWeb&referrerScenario=AddressBarCopied%2Eview

Azure OpenAI models fall into a few main families:

- GPT-4: A set of models that improve on GPT-3.5 and can understand as well as generate natural language and code.
- GPT-3.5: A set of models that improve on GPT-3 and can understand as well as generate natural language and code.
- Embeddings: A set of models that can convert text into numerical vector form to facilitate text similarity.
- DALL-E: A series of models that can generate original images from natural language.

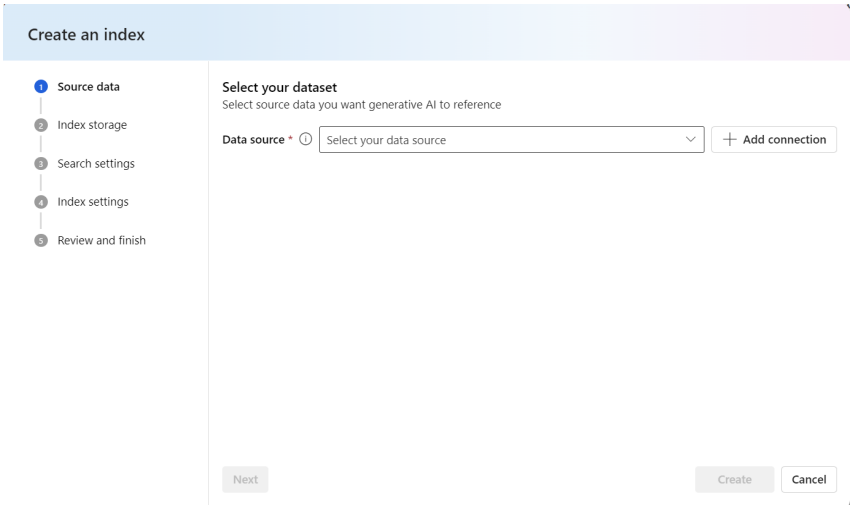
- **Azure OpenAI Studio:**

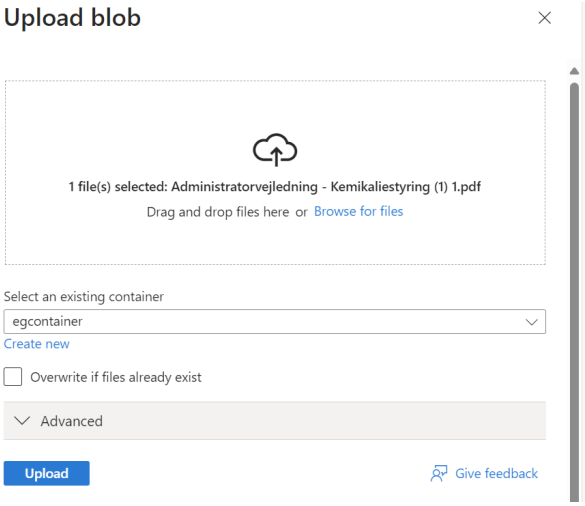
Click on deployment then click on create, lets go for real time endpoint as you can see all the LLM, other open-source models and GPT models. If you select a open source model like mixtral which is big model who's size is around 46.5 billion parameters and size s 186 GB so it ask for a virtual machine as you can see.mixtral 8x7b AI is the most comparable model in the open source ecosystem that is competes close enough to open AI GPT-35 model, but any way i will go for a GPT-35-turbo as it will cost token wise. After deployment we will get all the details along with API key so that we can use the deployments. You can test the deployment in this clicking open in playground option. Now i have my model deployed i want to perform retrieval augmented generation using my data which will be in the form of pdf,html etc.



So that i come to the indexes, Indexes uses azure AI search to create an index out of the document taht we will be uploading so in the background. Azure AI search take care of document creation storage and retrieval of document embedding, here i can select my data source from where y document lies. I have options to

connect Azure blob storage for upload file or folder here itself. Let me upload a folder that contains different type of documents with Danish and English text.





Home > Azure AI services | AI Search >

Create a search service ...

Basics Scale Networking Tags Review + create

Project details

Subscription * DEP_595_EGPL_DEV_AI

Resource Group * AI-Team

Create new

Instance Details

Service name * egaisearch

Location * North Europe

Pricing tier * **Standard**
25 GB/Partition, max 12 replicas, max 12 partitions, max 36 search units
Change Pricing Tier

Select Pricing Tier

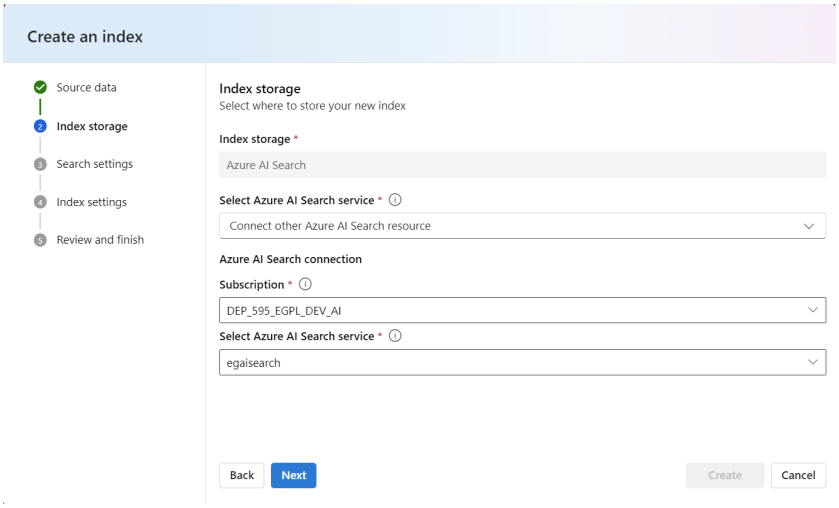
Browse available skus and their features

Sku	Offering	Indexes	Indexers	Vector storage	Total storage	Search units	Replicas	Partitions	Search unit cost/month (est.)
F	Free	3	3	25 MB	50 MB	1	1	1	$0.00
B	Basic	15	15	1 GB	2 GB	3	3	1	$75.14
S	Standard	50	50	3 GB/Partition	25 GB/Partition	36	12	12	$249.98
S2	Standard	200	200	12 GB/Partition	100 GB/Partition	36	12	12	$999.94
S3	Standard	200	200	36 GB/Partition	200 GB/Partition	36	12	12	$1,999.87
S3HD	High-density	1000	0	36 GB/Partition	200 GB/Partition	36	12	3	$1,999.87
L1	Storage Optimized	10	10	12 GB/Partition	1 TB/Partition	36	12	12	$2,856.22
L2	Storage Optimized	10	10	36 GB/Partition	2 TB/Partition	36	12	12	$5,711.69

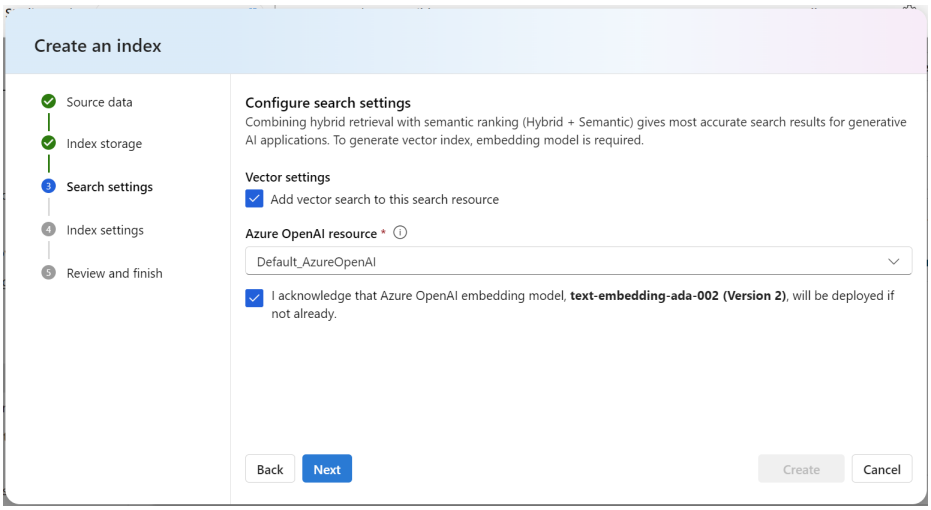
Select

Prices presented are estimates in your local currency that include only Azure infrastructure costs and any discounts for the subscription and location. The prices don't include any applicable software costs. Final charges will appear in your local currency in cost analysis and billing views. View Azure pricing calculator.

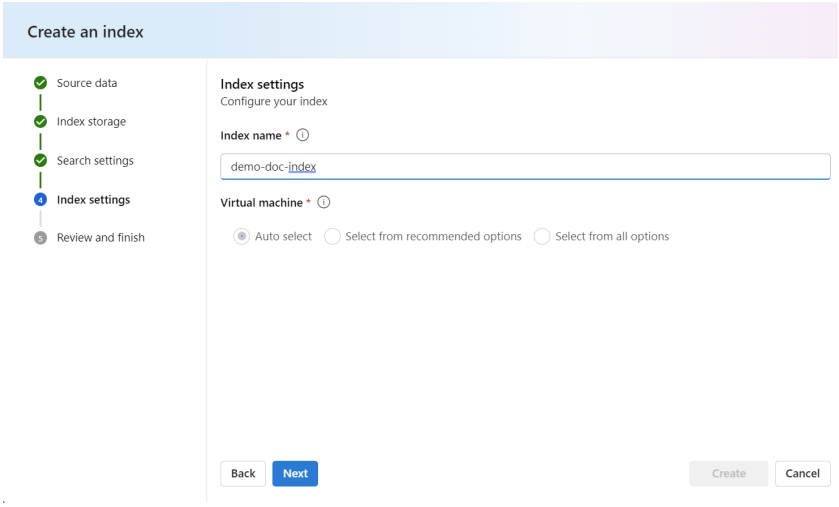
Then go to the index storage you will get a option of Azure AI search is simple. Azure AI search provides multiple pricing options based on the storage capacity required and read/writes replicas then click on next.

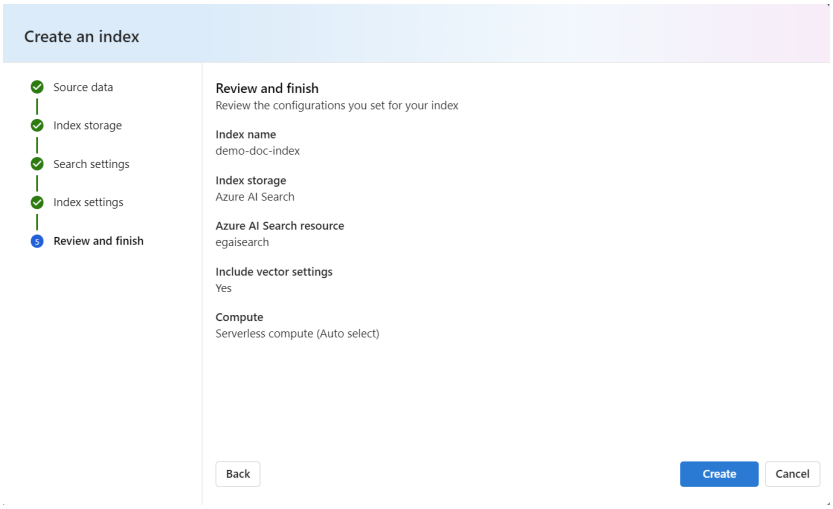


In the search setting s select vector search and enable the option for using embedding model, if not already deployed then click on next.



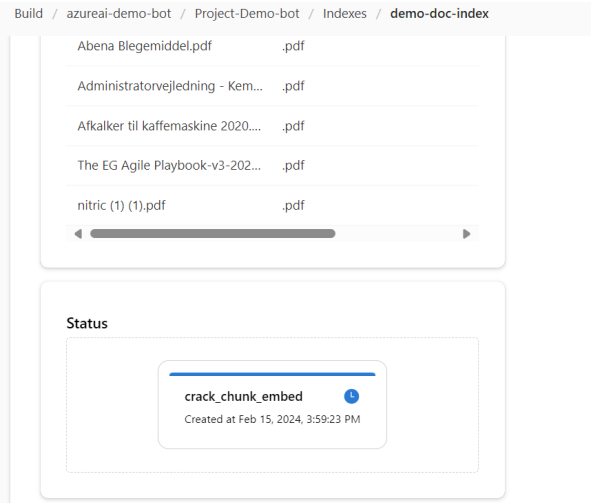
Let's got to the index setting give the index name and you will get all the details of index.

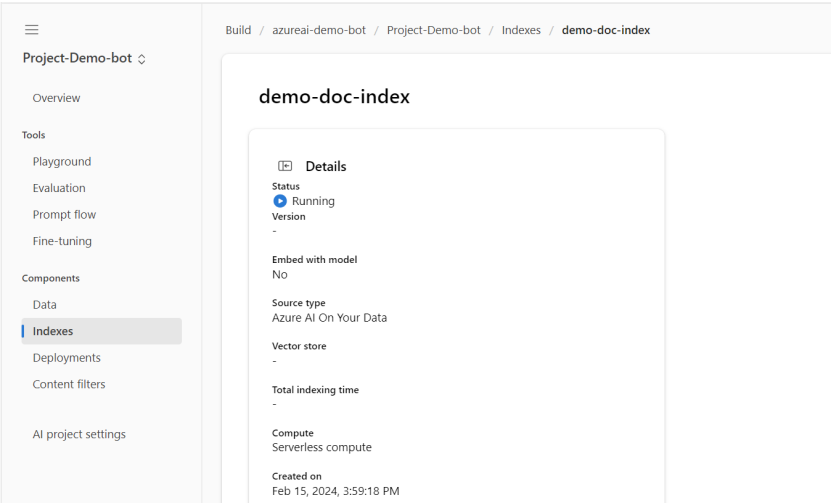




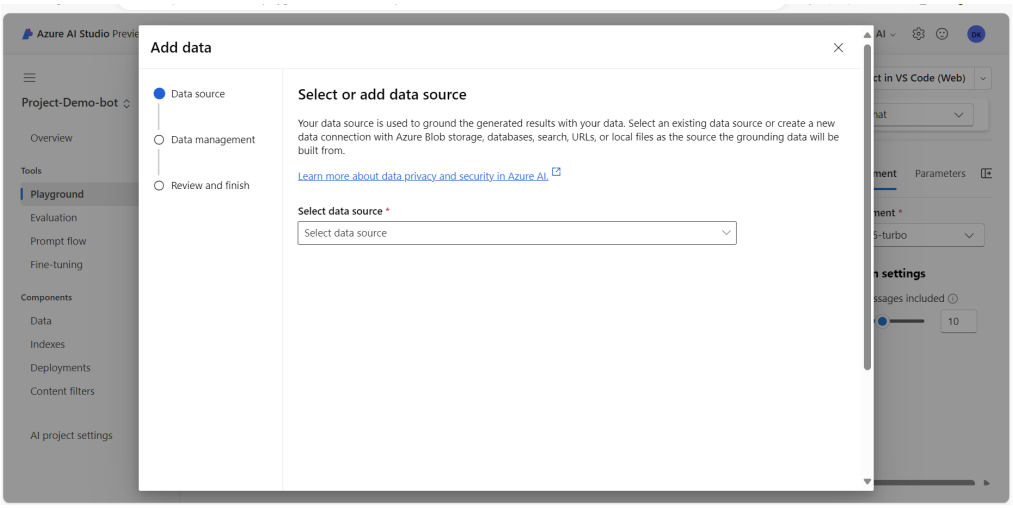
Once you created index then documents are processed to create embeddings. The time taken to process will depend on number of documents uploaded. Here it took about 3 minutes.

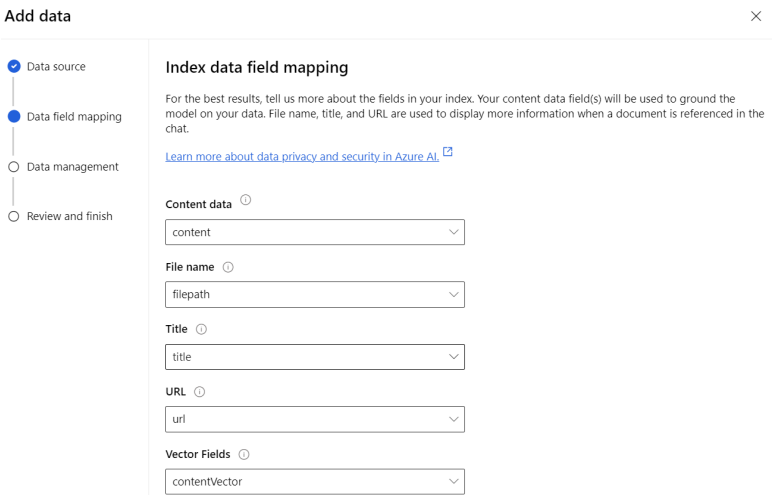
video by Devdas on creation of Azure AI chatbot based on your documentshttps://egonline-my.sharepoint.com/personal/presa_eg_dk/_layouts/15/stream.aspx?





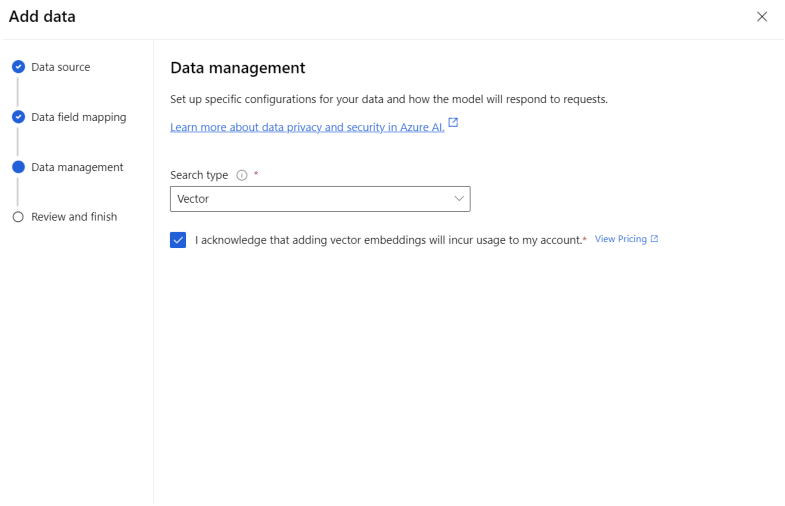
Now lets go to the playground.
click on add data source , then select data source. Here i am selecting ai search service and for search index i will select demo-doc-index which will point to index i created earlier. Then i check the option for data field mapping which will map my fields of document to retrieved references.



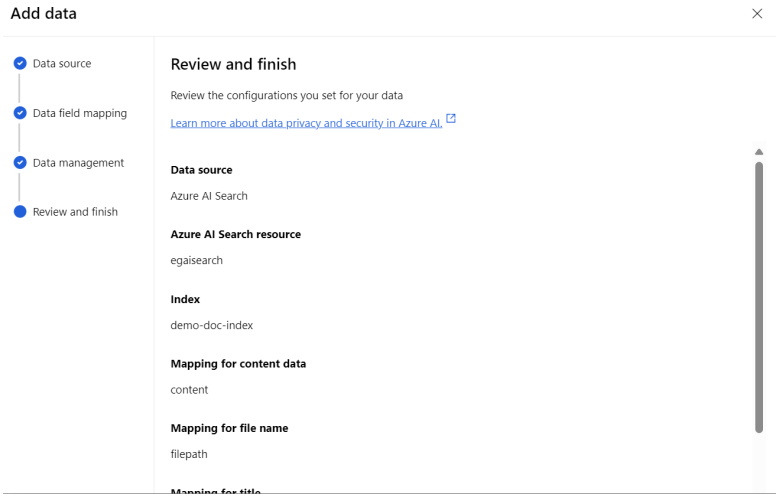


We acknowledge vector search option incurs charges.

video by Devdas on creation of Azure AI chatbot based on your documentshttps://egonline-my.sharepoint.com/personal/presa_eg_dk/_layouts/15/stream.aspx?



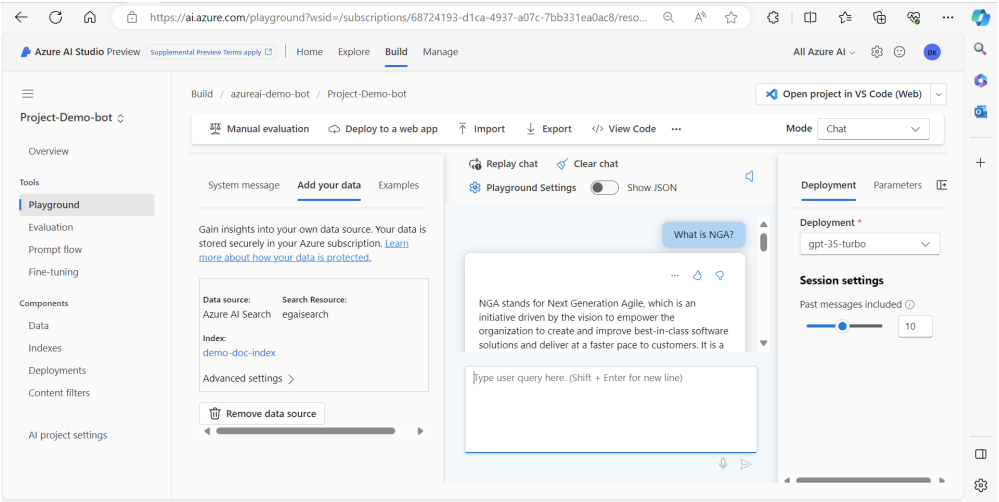
After that review and close.



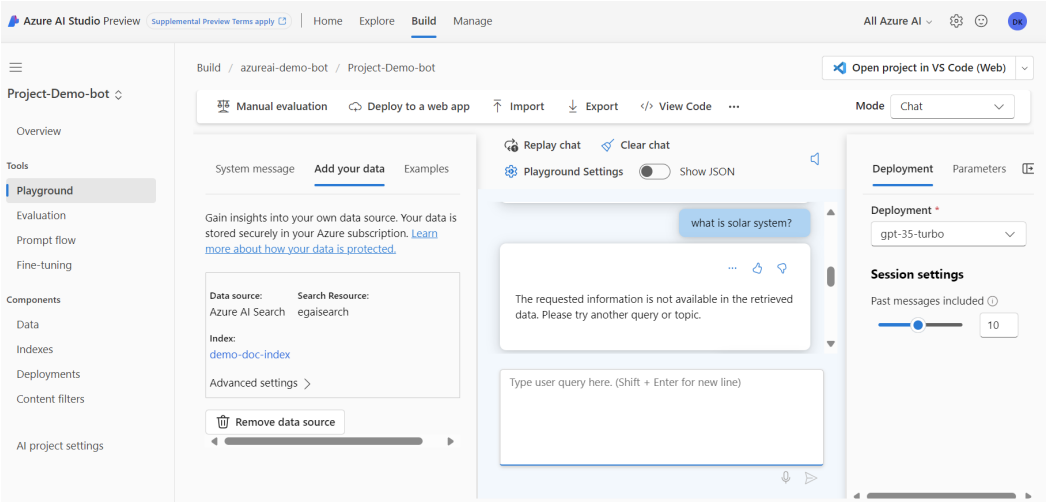
Now our chatbot is ready to test.
I will ask now what is NGA? it has given me answers based on the documents.

It also gives me references.

video by Devdas on creation of Azure AI chatbot based on your documentshttps://egonline-my.sharepoint.com/personal/presa_eg_dk/_layouts/15/stream.aspx?

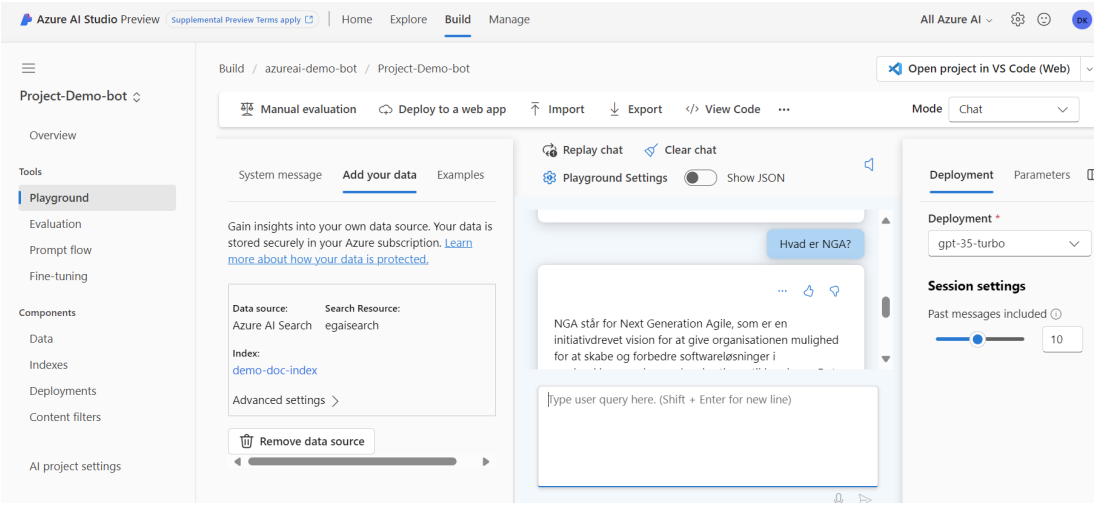


And if I ask question outside of document then it gives information not available in the document.



If i ask in Danish , it also gives answer in Danish.

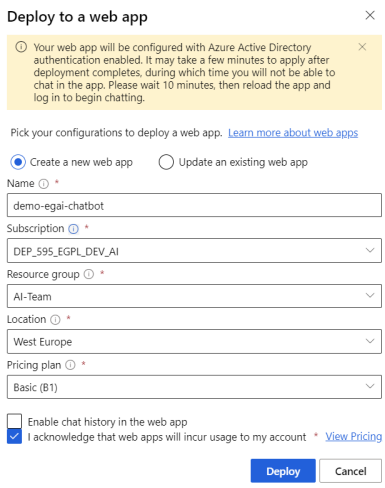
video by Devdas on creation of Azure AI chatbot based on your documentshttps://egonline-my.sharepoint.com/personal/presa_eg_dk/_layouts/15/stream.aspx?



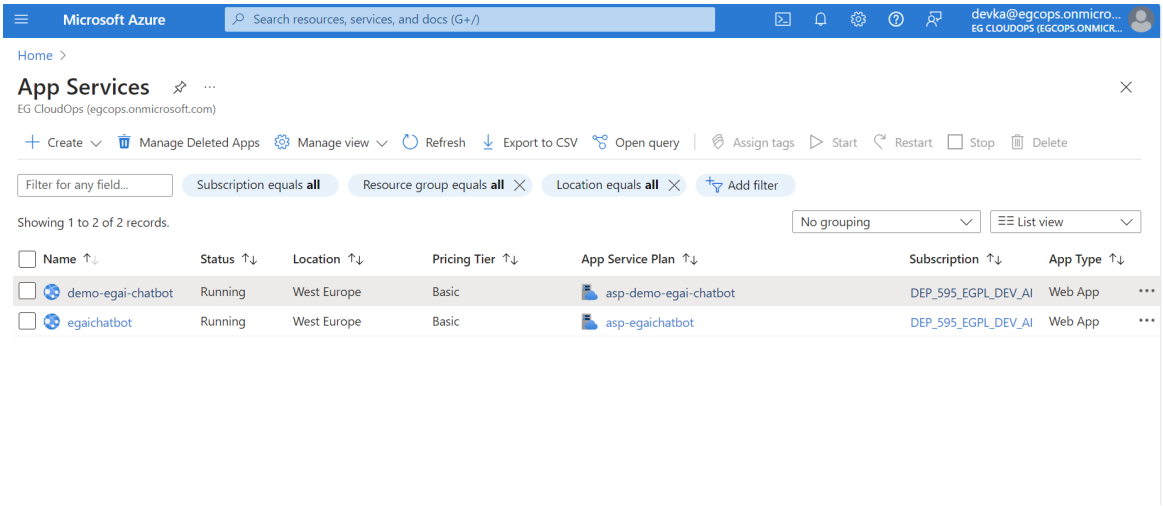
Now our application is ready to deploy.

Let's click on deploy to web app then fill the details of application.

If you want the chat history to be saved we can enable the option and our chat history will be stored in Azure Cosmos DB, then click on Deploy.

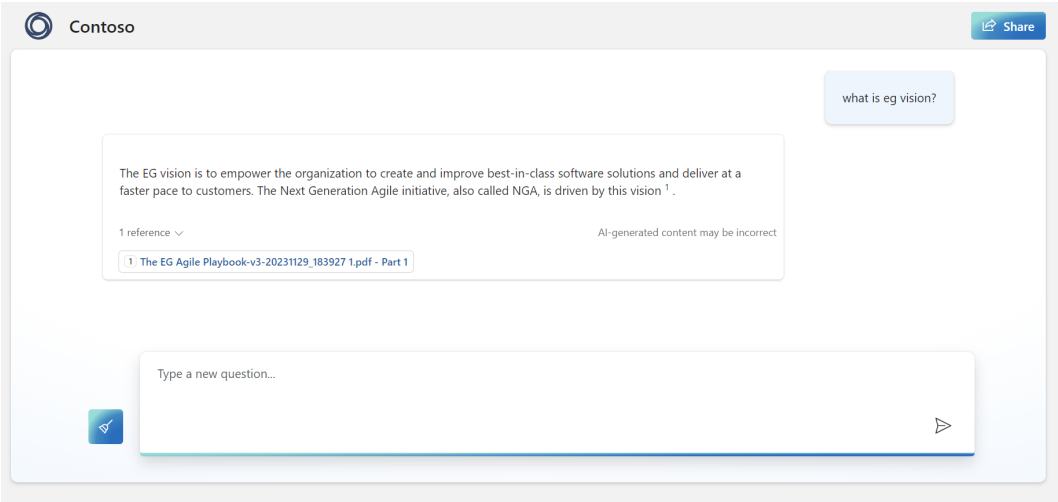


Now the app will be deployed, lets go to the azure app service and open the url.



So this is the deployed azure chatbot openai.

I am able to login this chatbot app with azure account credentials in my browser through azure sso. However company wide we need to configure authentication accordingly.



You can Add your chatbot with various channels-

Go to the Bot services and create the bot,

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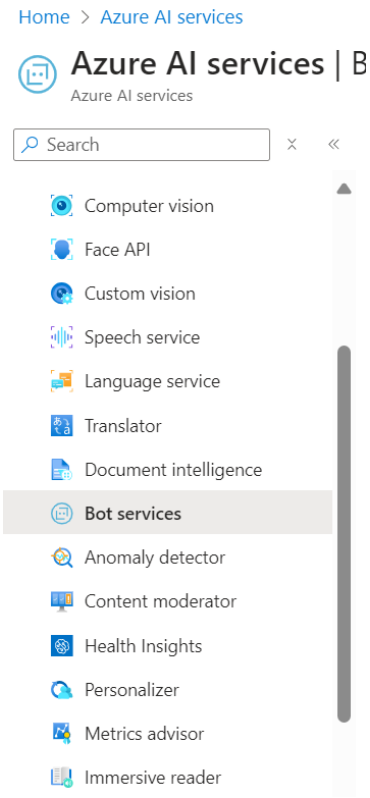
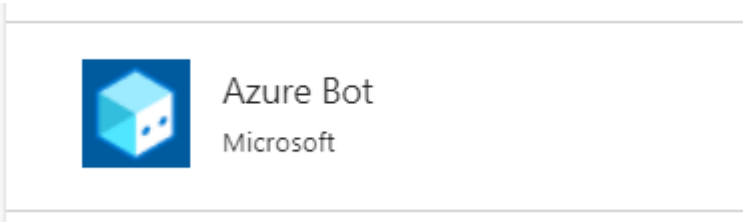
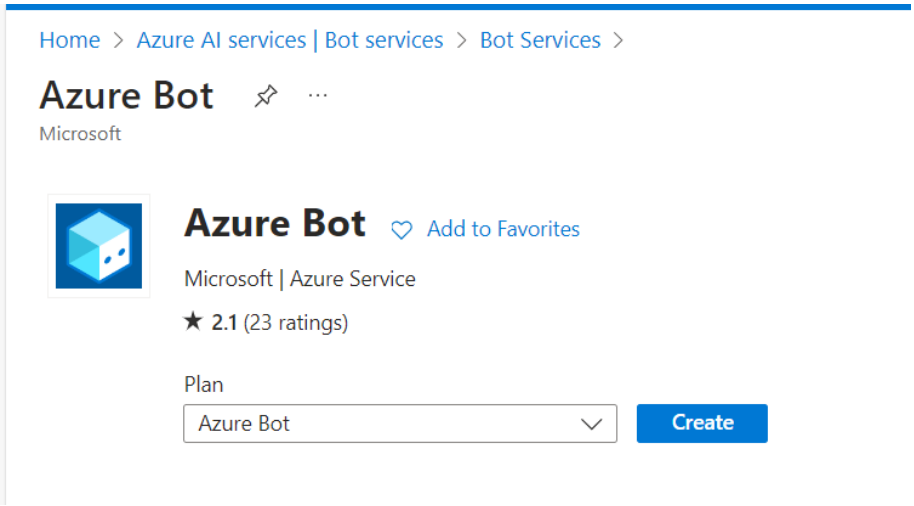


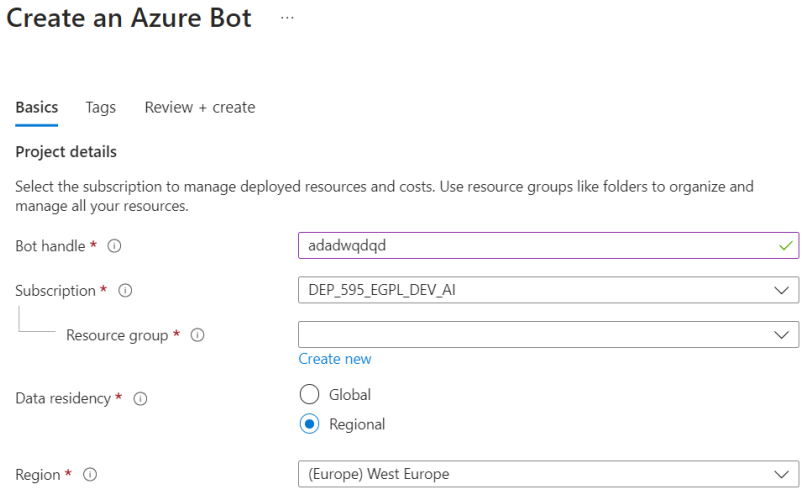
Create the Azure bot



please click on create



Fill the correct information



select the type of app is User-Assigned managed identity and create the bot

Microsoft App ID

A Microsoft App ID is required to create an Azure Bot resource. If your bot app doesn't need to access resources outside of its home tenant and if your bot app will be hosted on an Azure resource that supports Managed Identities, then choose option User-Assigned Managed Identity so that Azure takes care of managing the App credentials for you. Otherwise, depending on whether your bot will be accessing resources only in it's home tenant or not, choose either Single tenant or Multi tenant option respectively.

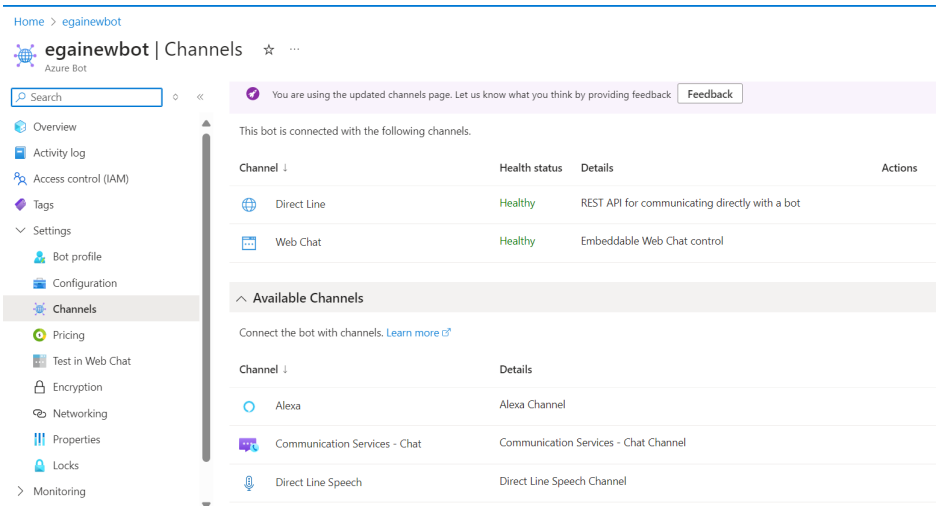
Type of App: User-Assigned Managed Identity

Note: For User-Assigned Managed Identity and Single Tenant app, Composer is not yet supported for bots with these app types. BotFramework SDK (C# or Javascript) version 4.15.0 or higher is needed for these app types.

A User-assigned managed identity can be automatically created below or you can manually create your own, then return to input your new App ID, tenant ID and MSI resource ID in the open fields.
Manually create a User Managed Identity

Creation type
- (●) Create new Microsoft App ID
- ( ) Use existing app registration

After creation go to the resources and under the settings click on channels



Available channels

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Alexa	Alexa Channel
Communication Services - Chat	Communication Services - Chat Channel
Direct Line Speech	Direct Line Speech Channel
Email	O365 Email Channel
Facebook	Support for Text Messaging via Facebook
GroupMe	GroupMe Channel
LINE	Support for LINE Channel
Microsoft Teams	Microsoft Teams Channel
Omnichannel	Omnichannel Channel
Slack	Slack Channel
Telegram	Telegram Channel

The bot is connected with the following channels

You are using the updated channels page. Let us know what you think by providing feedback Feedback

This bot is connected with the following channels.

Channel ↓	Health status	Details	Actions
Direct Line	Healthy	REST API for communicating directly with a bot	
Web Chat	Healthy	Embeddable Web Chat control	

click on the web chat and click on the default site

Configure communication between your bot and end users with Web Chat. The following instructions will help with how to connect a bot to the Web Chat channel. Learn more

Sites

Configure the sites that bind with the bot for Web Chat channel.

Site name	Enabled ↓	Authentication method	Preview controls	Actions
Default Site	Yes	Secrets	Enable Preview	...

After you will get the Embedded code

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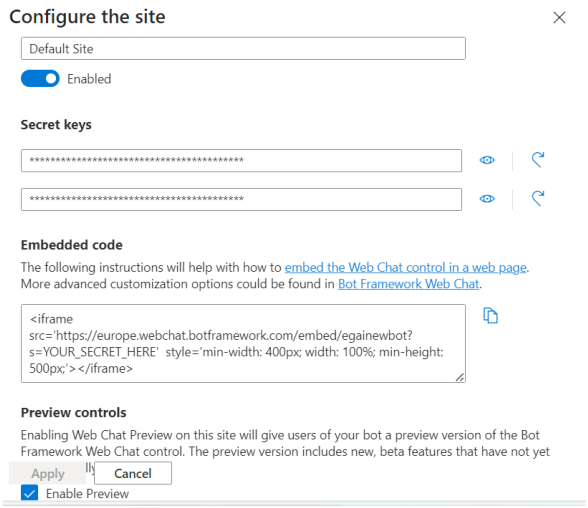

Certainly! Let's compare **Azure OpenAI Chatbot** (ChatGPT) and **Microsoft Copilot Chatbot** to understand their differences:

1. **Azure OpenAI Chatbot (ChatGPT)**:
    - **Pros**:
        - **Advanced AI**: ChatGPT leverages GPT-3.5 and GPT-4o models, offering powerful capabilities for free users.
        - **Multimodal**: It can process text, images, and sounds using GPT-4o.
        - **Free Access**: Available for free with a registered account.
        - **Popular**: Widespread preview since November 2022[4][5].
    - **Cons**:
        - **Limited Flexibility**: Primarily trained for English and French, restricting language variety.
        - **Free Version Limitations**: Paying for the Plus version unlocks additional features.
        - **File upload limit**: We can upload files up to 16 MB in size.

Azure OpenAI available models-

References-

Azure bot creation-Create an Azure Bot resource in the Azure portal - Bot Service | Microsoft Learn[6]

https://learn.microsoft.com/en-us/azure/communication-services/quickstarts/chat/quickstart-botframework-integration

4 https://www.zdnet.com/article/chatgpt-vs-microsoft-copilot-vs-gemini-which-is-the-best-ai-chatbot/
5 https://www.zdnet.com/article/chatgpt-vs-microsoft-copilot-vs-gemini-which-is-the-best-ai-chatbot/
6 https://learn.microsoft.com/en-us/azure/bot-service/abs-quickstart?view=azure-bot-service-4.0&wt.mc_id=searchAPI_azureportal_inproduct_rmskilling&sessionId=eb73102135b34087bdb14002740f3531&tabs=multitenant

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https://learn.microsoft.com/en-us/azure/bot-service/bot-service-channel-connect-webchat?view=azure-bot-service-4.0